Full Development Flow — From Zero To Live Product

Phase 0 — Setup (Week 1)

```
DAY 1 — Development Environment
━━━
```

```
│   ├── pdf/
│   ├── qr/
│   ├── admin/
│   └── licence/
└── shared/
    ├── widgets/
    ├── theme/
    └── constants/
```

DAY 3 — Add All Packages
━━━

```
  connectivity_plus: latest

  # Device info
  device_info_plus: latest
  network_info_plus: latest

  # File system
  path_provider: latest

  # Charts (admin)
  fl_chart: latest

  # Local server
  shelf: latest
  shelf_router: latest

  # AI
  tflite_flutter: latest
  palette_generator: latest

  # Kiosk mode
  window_manager: latest
  launch_at_startup: latest

  # Utils
  intl: latest
  uuid: latest
  crypto: latest
  equatable: latest
```

DAY 4 — Design System Setup
━━

shared/theme/app_theme.dart

→ Brand colors
→ Typography
→ Button styles
→ Card styles
→ Spacing constants
→ Border radius
→ Shadow styles

DAY 5 — Git Repository
━━

→ Create private GitHub repo
→ Setup .gitignore
→ Create branches:
  main (production)
  develop (active development)
  feature/* (each feature)
→ First commit
```

---

## Phase 1 — Database Layer (Week 2)

```
STEP 1 — Define All Models
━━━

catalogues_table
→ id, brand_id, name
→ category, display_order
→ thumb_path, is_active

laminates_table
→ id, catalogue_id, code, name
→ texture_path, thumb_path
→ color_primary, color_secondary
→ pattern_type, finish_type
→ warmth, style_tags, is_glossy
→ is_active, display_order

designs_table
→ id, staff_id, customer_name
→ room_id, screenshot_path
→ zones_data (JSON)
→ pdf_path, created_at

staff_table
→ id, name, pin_hash
→ role, is_active
→ created_at

analytics_table
→ id, laminate_id, room_id
→ action_type, staff_id
→ created_at

STEP 3 — Database Service
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━

core/services/database_service.dart

Methods needed:
→ getRoomsByCategory()
→ getCatalogues()
→ getLaminatesByCatalogue()
→ searchLaminates(query)
→ saveDesign(design)
→ getDesigns()
→ logAnalytics(event)

STEP 4 — Seed Initial Data
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━

Create seed script
Populates database from:
→ JSON file with all laminate data
→ Room configuration file
Runs once on first install
```

Phase 2 — Canvas Engine (Week 3-4)

```
THIS IS THE MOST CRITICAL PHASE
Get this right before anything else

STEP 1 — Basic Canvas Setup
━━━

```
canvas.drawImage(
  shadowImage,
  Offset.zero,
  Paint()..blendMode = BlendMode.multiply
);

Layer 4: AO overlay
canvas.drawImage(
  aoImage,
  Offset.zero,
  Paint()
    ..blendMode = BlendMode.multiply
    ..color = Color.fromRGBO(0,0,0,0.5)
);
```

STEP 3 — Mask System
━━

Load mask PNG as image
Convert to Path using pixel data
White pixels = inside clip path
Black pixels = outside
This path used for clipPath()

STEP 4 — Perspective Transform
━━

Floor needs perspective
Texture must follow vanishing point

Store 4 corner points per zone
in database (set during room setup)

top-left, top-right,
bottom-left, bottom-right

Calculate Matrix4 from these points
Apply to ImageShader matrix
Texture warps to match floor angle

STEP 5 — Texture Scaling
━━

Real laminate = 60cm × 60cm
Room photo represents real space
Calculate pixels per meter in photo
Scale texture to match real world

STEP 6 — Texture Swap
━━━

```
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
features/login/login_screen.dart

Two options:
→ Staff PIN login
→ Guest/Customer mode

PIN Login:
→ Dropdown select staff name
→ Number pad widget
→ 4 digit PIN
→ Hash and compare with SQLite
→ Set session in Riverpod

Guest Mode:
→ Skip login
→ No design saving
→ PDF still works
→ No analytics tracking

Session Management:
→ Auto logout after 30 min idle
→ Touch activity resets timer
→ Background timer in service
```

Phase 4 — Home + Room Picker (Week 6)

```
HOME SCREEN
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
features/home/home_screen.dart

Category grid:
→ 6 category cards
→ Each card:
  Full bleed image background
  Category name overlay
  Hover/touch scale animation
→ flutter_animate entrance animation
→ Staggered card appearance
```

On category tap:
→ Navigate to room picker
→ Pass category as parameter

ROOM PICKER SCREEN
━━━

ROOM CANVAS WIDGET

features/visualizer/room_canvas_widget.dart

→ CustomPaint with RoomPainter
→ RepaintBoundary wrapper
  (for screenshot capture)
→ DragTarget overlay
  (invisible, covers entire canvas)
→ Zone highlight overlay
  (glows when laminate dragged over)
→ Active zone indicator
  (shows which zone is selected)

CATALOGUE PANEL WIDGET

features/catalogue/catalogue_panel.dart

Top section:
→ Catalogue tab bar
  (Natural Wood, Marble, Solid...)
→ Search bar

Middle section:
→ GridView.builder (lazy)
→ 3 columns
→ Each card:
  Laminate thumbnail
  Name (truncated)
  Code
  Draggable wrapper

Bottom section:
→ Active selection summary
  Floor: GL1034 Oak Natural
  Wall: GL2089 Marble White

DRAGGABLE LAMINATE CARD

```
Draggable<LaminateModel>(
  data: laminate,

  // What shows while dragging
```

```
  feedback: ScaleTransition(
    scale: 1.2,
    child: LaminateCardDragging(),
  ),

  // Original card fades
  childWhenDragging: Opacity(
    opacity: 0.4,
    child: LaminateCard(),
  ),

  onDragStarted: () {
    // Highlight drop zones
    ref.read(showZoneHighlight.notifier)
        .state = true;
    // Load full texture in background
    textureService.preload(laminate.id);
  },

  onDragEnd: (_) {
    ref.read(showZoneHighlight.notifier)
        .state = false;
  },

  child: LaminateCard(),
)
```

DRAG TARGET ZONES
━━

Each zone (floor, wall) is a DragTarget
Positioned over canvas using Stack

```
DragTarget<LaminateModel>(
  onAccept: (laminate) {
    // Update zone texture
    ref.read(zoneTextureProvider(zoneId)
        .notifier)
        .updateTexture(laminate);

    // Log analytics
    analyticsService.log(
      laminateId: laminate.id,
      action: 'dropped',
      zone: zoneId,
```

```
    );

    // Play success animation
    zoneSuccessAnimation.play();
  },

  builder: (context, candidates, rejected) {
    final isHovering = candidates.isNotEmpty;
    return ZoneHighlightOverlay(
      isActive: isHovering,
      // Glowing border when hovering
    );
  },
)

TOP ACTION BAR
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
→ Back button
→ Room name
→ Save Design button
→ Compare button
→ Generate PDF button
→ Reset button
```

Phase 6 — Asset Loading Service (Week 9)

```
TEXTURE CACHE SERVICE
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
core/services/texture_service.dart

LRU Cache:
→ Max 30 textures in memory
→ Least recently used auto removed
→ New texture loads on demand

Methods:
→ loadTexture(laminateId) → ui.Image
→ preloadTexture(laminateId)
→ preloadNext10(currentIndex)
→ clearCache()
```

Loading priority:
1. Visible thumbnails first (tiny)
2. Currently hovered laminate
3. Next 10 in scroll direction
4. Rest on demand

ROOM ASSET SERVICE
━━━

Modern/Traditional/Rustic/Luxury

Step 4: Match laminates
SQLite query:
SELECT * FROM laminates
WHERE color_distance(color_primary,
    userPrimaryColor) < threshold
AND style_tags CONTAINS detectedStyle
ORDER BY match_score DESC
LIMIT 12

Step 5: Return recommendations
Show in catalogue panel
Highlight match percentage
```

---

## Phase 8 — Save + Compare (Week 11)

```
SAVE DESIGN
━━
User taps Save:
→ Show name input dialog (optional)
→ Capture canvas screenshot
  using RepaintBoundary
→ Save screenshot to app documents
→ Save design record to SQLite:
  room_id, staff_id,
  floor_laminate_id,
  wall_laminate_id,
  screenshot_path,
  customer_name,
  created_at
→ Show success toast

COMPARE MODE
━━
User taps Compare:
→ Show saved designs list
→ Select 2 designs
→ Split screen layout
  Left: Design 1
```
```

```
  Right: Design 2
→ Both rendered with CustomPainter
→ Laminate details below each
→ Option to select either as final
```

Phase 9 — PDF Generation (Week 12)

```
PDF SERVICE
━━━
```

→ Company address + website

Save PDF:
→ path_provider gets documents dir
→ Save as design_[timestamp].pdf
→ Return file path

Show preview:
→ printing package
→ Full screen PDF preview
→ Option to close
```

---

## Phase 10 — QR + Phone Download (Week 12)

```
LOCAL SERVER SERVICE
━━
core/services/local_server_service.dart

Using shelf package:
→ Start HTTP server on port 8080
→ Serve PDF file at unique URL
  /download/[unique_id]
→ Get device local IP
  using network_info_plus
→ Build URL:
  http://[localIP]:8080/download/[id]

QR SCREEN
━━
features/qr/qr_screen.dart

→ Generate QR from URL
  using qr_flutter
→ Display large on screen
→ Show instructions:
  "Connect to [WiFi Name]
   Scan to download your design"
→ Countdown timer 10 minutes
→ Auto stop server after timer
→ WhatsApp share button
```
```

```
  (opens WhatsApp with PDF attached)
→ Start New Design button
```

Phase 11 — Licence System (Week 13)

```
LICENCE SERVICE
━━━
```

→ Activate button
→ Contact support link
→ Error messages:
  Invalid key
  Already activated
  Expired
  No internet
```

---

## Phase 12 — Admin Dashboard (Week 14)

```
ADMIN SCREEN
━━━

→ Add staff member
→ Set PIN
→ View activity log
→ Deactivate staff

SETTINGS
→ Brand name
→ Contact details for PDF
→ WiFi network name
→ Auto logout timer
→ Screen timeout
→ App version info

DATA EXPORT
→ Export designs CSV
→ Export analytics PDF report
→ Backup database to USB
```

Phase 13 — Kiosk Mode + Windows Setup (Week 15)

```
KIOSK CONFIGURATION
━━━

```
  );

  // Auto start on Windows boot
  await launchAtStartup.setup(
    appName: 'Laminate Visualizer',
    appPath: Platform.resolvedExecutable,
  );
  await launchAtStartup.enable();

  runApp(
    ProviderScope(child: App()),
  );
}

WINDOWS KIOSK MODE
━━━
```

→ Color matching algorithm
→ PDF generation
→ Asset loading service

WIDGET TESTS
━━━

CREATE INSTALLER
━━━

```
│   └── auth.js
└── database/
    └── postgres/
```

Deploy on Railway or Render:
→ Free tier handles 100+ clients
→ Auto SSL certificate
→ Custom domain
→ ₹0 – ₹1,500/month

ADMIN DASHBOARD (Web)
━━

Simple React web app:
→ Your internal tool only
→ Generate licence keys
→ View all activations
→ Revoke licences
→ See heartbeat status
→ Payment status per client
```

Complete Timeline Summary
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━

| WEEK | PHASE | OUTPUT |
|---|---|---|
| 1 | Setup + Structure | Project ready |
| 2 | Database layer | Data working |
| 3-4 | Canvas engine | Swap working |
| 5 | Splash + Login | Auth working |
| 6 | Home + Room picker | Navigation done |
| 7-9 | Main visualizer | Core feature done |
| 9 | Asset loading | Performance good |
| 10 | AI matching | Smart recommend |
| 11 | Save + Compare | Memory feature |
| 12 | PDF + QR | Output working |
| 13 | Licence system | Protection done |
| 14 | Admin dashboard | Management done |
| 15 | Kiosk + Windows | Deploy ready |
| 15-16 | Testing | Bugs fixed |
| 16 | Build + Installer | .exe ready |

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TOTAL: 16 weeks = 4 months
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
```

Team Needed

ROLE	TASK	COST/MONTH
Flutter Dev	Canvas + Core	₹1,00,000
UI Dev	Screens + Design	₹50,000
AI Engineer	TFLite + Colors	₹80,000
Content Artist	Assets + Masks	₹30,000
QA Engineer	Testing	₹30,000
Node.js Dev (part time)	Licence server	₹40,000
Total per month		₹3,30,000
Total 4 months		₹13,20,000

One Rule Above Everything
Build the canvas engine first.
Test it on real hardware early.
If shadows look real on week 4
everything else is just screens.
If shadows look fake on week 4
stop and fix before building anything else.

The entire product's value
is in that one moment
when customer drops a laminate
and says "oh wow yeh toh real lag raha hai"

That moment is worth
every rupee and every week
you invest in getting it right.

Simple Real Product Setup Flow

Step 1 — What You Need To Collect First

```
FROM THE BRAND (Before anything)
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
→ Physical sample book (courier to you)
→ Catalogue PDF (all laminate names + codes)
→ Brand logo (AI or PNG high res)
→ Brand colors (hex codes)
→ Brand fonts
→ Room photos if they have any
→ Excel sheet of all laminate codes + names
```

Step 2 — Room Images

How Many You Need

```
Per category minimum:
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
Living Room    → 4 to 6 rooms
Bedroom        → 4 to 6 rooms
Kitchen        → 3 to 4 rooms
Office         → 3 to 4 rooms
Commercial     → 2 to 3 rooms
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
Total          → 16 to 23 rooms
For a solid launch: 20 rooms
```

Rules For Room Photos

```
MUST HAVE
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
→ Camera perfectly straight
  (no tilt, no fisheye)
→ Floor clearly visible
  minimum 40% of image
→ Even flat lighting
  no harsh shadows from windows
→ Neutral base surfaces
  plain floors and walls
  (laminate will replace these)
→ Minimum 4000 × 3000 px resolution
```

→ Shot in RAW format
→ No people in photo
→ Furniture can stay
  (adds realism)

AVOID
━━━

Step 3 — Process Each Room Photo

What You Create From Each Photo

```
For every single room photo
you need to create these files:

1. base.webp
   The original room photo
   Compressed to WebP
   Resolution: 1920 × 1080
   File size: 300 – 500KB

2. thumb.webp
   Small preview for room picker
   Resolution: 400 × 225
   File size: 20 – 40KB

3. mask-floor.png
   White = floor area
   Black = everything else
   Same resolution as base
   File size: 50 – 100KB

4. mask-wall.png
   White = wall area
   Black = everything else
   Same resolution as base
   File size: 50 – 100KB

5. shadow.png
   Baked shadows and highlights
   Exported from original photo
   Same resolution as base
   Multiply blend mode on top
   File size: 100 – 200KB

6. ao.png
   Ambient occlusion
   Dark corners and edges only
   Low opacity multiply on top
   File size: 50 – 100KB
```

Total files per room = 6 files
20 rooms = 120 files total
```

### How To Create Mask Files
```
STEP 1
Open room photo in Photoshop

STEP 2
Use SAM (Segment Anything Model)
or Photoshop Quick Select tool
Click on floor area
AI auto selects entire floor

STEP 3
Refine edges
Especially around furniture legs
Rug edges
Skirting boards

STEP 4
Fill selection with white
Everything else black
Export as mask-floor.png

STEP 5
Repeat for wall area
Export as mask-wall.png

STEP 6
Create shadow layer:
Duplicate original photo
Convert to grayscale
Increase contrast
Darken shadow areas
Keep only shadow information
Export as shadow.png

Time per room: 45 minutes to 1.5 hours
With SAM tool: 20 to 30 minutes
20 rooms total: 2 to 3 days work
```
```

Step 4 — Laminate Textures

How Many You Need

```
For a solid launch per brand:
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
Catalogue 1 (Natural Wood)   → 80 to 120
Catalogue 2 (Marble/Stone)   → 60 to 80
Catalogue 3 (Solid Colors)   → 40 to 60
Catalogue 4 (Abstract)       → 40 to 60
Catalogue 5 (Fabric/Textile) → 30 to 50
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
Total launch                 → 250 to 370
Good number to start: 300 laminates

Full catalogue later: 500 to 1500
Add in monthly batches
```

What You Need Per Laminate

```
For every single laminate you need:

1. texture.webp
   Seamless tileable texture
   Resolution: 1024 × 1024 px
   File size: 80 – 150KB

2. thumb.webp
   Small catalogue card image
   Resolution: 200 × 200 px
   File size: 8 – 15KB

3. Database entry
   Code: GL1034
   Name: Oak Natural Sunrise
   Catalogue: Natural Wood
   Category: Wood
   Primary color: #8B6340
   Secondary color: #C4A882
   Pattern: Wood grain
   Finish: Matte
```

```
  Warmth: Warm
  Style tags: Modern, Classic
  Is glossy: No

Total per laminate = 2 files + 1 DB entry
300 laminates = 600 files + 300 DB entries
```

How To Create Laminate Textures

```
SCANNING PROCESS
━━━
```

Step 5 — Database Setup

SQLite Structure Simple

```
ROOMS TABLE
──────────────────────────────────────────────────────────
id, category, style, name,
base_path, thumb_path,
shadow_path, ao_path,
has_floor_zone, has_wall_zone

ZONES TABLE
──────────────────────────────────────────────────────────
id, room_id, zone_type,
mask_path, perspective_points

CATALOGUES TABLE
──────────────────────────────────────────────────────────
id, name, category,
display_order, thumb_path

LAMINATES TABLE
──────────────────────────────────────────────────────────
id, catalogue_id, code, name,
texture_path, thumb_path,
color_primary, color_secondary,
pattern_type, finish_type,
warmth, style_tags, is_glossy
```

Step 6 — App Asset Folder Structure

```
assets/
│
├── rooms/
│   ├── living-room/
│   │   ├── modern/
│   │   │   ├── base.webp
│   │   │   ├── thumb.webp
│   │   │   ├── mask-floor.png
│   │   │   ├── mask-wall.png
```

```
│   │   │   ├── shadow.png
│   │   │   └── ao.png
│   │   ├── classic/
│   │   ├── luxury/
│   │   └── minimal/
│   ├── bedroom/
│   ├── kitchen/
│   └── office/
│
├── laminates/
│   ├── natural-wood/
│   │   ├── GL1001-texture.webp
│   │   ├── GL1001-thumb.webp
│   │   ├── GL1002-texture.webp
│   │   └── GL1002-thumb.webp
│   ├── marble/
│   ├── solid/
│   └── abstract/
│
└── database/
    └── catalogue.db
```

Step 7 — Complete Asset Count Summary

```
────────────────────────────────────────────────────────────────────────────────
ASSET                COUNT     SIZE      COST
────────────────────────────────────────────────────────────────────────────────
Room base images     20        10MB      ₹20,000
Room thumbnails      20        1MB       included
Floor masks          20        2MB       ₹6,000
Wall masks           20        2MB       included
Shadow layers        20        4MB       ₹4,000
AO layers            20        2MB       included
Laminate textures    300       40MB      ₹10,000
Laminate thumbs      300       5MB       included
SQLite database      1         2MB       ₹5,000
────────────────────────────────────────────────────────────────────────────────
```

TOTAL FILES 721 68MB
TOTAL COST ₹45,000
━━━

→ Then licence system

MONTH 4
━━━